School of Computer Science Engineering and Technology

Lab-01

Natural Language Processing

Course- B.Tech.	**Type-** Specialization/ NSPL Elective - I
Course Code- CSET346	**Course Name-** Natural Language Processing (Lab)
Date- 17/08/2026 – 21/08/2026	**Semester-** ODD

CO Mapping

Q(s)	CO1	CO2	CO3
Q1 and Q2		√	√

Objective:

Students will be able:

- To understand how to use different file formats in Colab / Kaggle platform.
- To use the text data handing in python environment.

LAB ASSIGNMENT- 1

Task 1.

Getting familiar with the environment (Colab/Kaggle/Git), popular NLP libraries like nltk, re, spaCy, Data types and Visualization

Step 1: Download the dataset files belong to the following data formats# from internet. The files may belong to any dataset available online.

Step 2: Read these files inside the python code. Some of the file formats cannot be read using default python packages. In this case, explore the python packages suitable for reading the files.

Step 3: Print the properties of the data files such as size, shape, dimensions, etc.

Step 4: Visualize each of these data files using graphs, diagrams, etc.

- Table data visualization: line graph, bar graph, histogram chart, pie chart, scatter plot
- Image visualization: image plot, 3d plot
- Video visualization: video player
- Audio visualization: audio player, spectrogram
- Text visualization: Word cloud, bubble cloud (some more in http://vallandingham.me/textvis-talk/)

Data Formats

1. Tabular, Spreadsheet and Interchange Data Formats

- "Table" — generic tabular data (.dat), "CSV" — comma-separated values (.csv), "TSV" — tab-separated values (.tsv), “ARFF” - Attribute-Relation File Format (.arff) – Read and visualize the data
- "XLS" — Excel spreadsheet (.xls), "XLSX" — Excel 2007 format (.xlsx), "ODS" — OpenDocument spreadsheet (.ods), "SXC" — OpenOffice 1.0 spreadsheet file (.sxc), "DIF" — VisiCalc data interchange format (.dif) – Read and visualize the data
- "JSON" — JavaScript Object Notation (.json), "UBJSON" — Universal Binary JSON (.ubj), “HTML” – Hypertext Markup Language (.html), “XML” - eXtensible Markup Language (.xml) - Read and Parse the data

2. Data File Formats

- PKL – Pickle format, HDF5, Zip, SQL, MAT, NPY, NPZ – Read and display the data

3. Image Data Formats

- JPG, PNG, BMP, TIFF – Read and display the image
- 3D medical Images: DICOM, MHA – Read and display the image

4. Video Data Formats

- MP4, AVI, MPEG – Read and play the video

5. Audio Data Formats

- MP3, MIDI, WAV – Read and play the audio

6. Text Data Formats

- TXT, PDF, DOC – Read and parse the data

Suggested Platform: Python: Azure Notebook/Google Colab Notebook/ Kaggle, packages such as Numpy, Pandas, Sklearn, matplotlib.

Task 2:
You are provided with an uncleaned paragraph. Your task is to perform a series of data cleaning steps to preprocess the text and make it ready for NLP tasks. You will explore the utility of **re** package- regular expression package.

Objective: The objective of this assignment is to familiarize with the essential data cleaning steps in Natural Language Processing (NLP). Students will work with a challenging uncleaned paragraph, applying various data cleaning techniques to preprocess the text and make it suitable for further NLP tasks. Also, students will learn how to tokenize text data, apply stemming to reduce words to their root form, and perform lemmatization to obtain meaningful base forms using dictionary-based methods. These foundational operations are essential for preprocessing text in any NLP application.

Data Cleaning Steps:

1. Lowercasing: Convert the entire paragraph to lowercase to ensure consistent capitalization.
2. Removing Extra Whitespace: Remove extra spaces and ensure that only a single space separates word.
3. Handling Contractions: Correct contractions to their full forms (e.g., "I <3 nlp" to "I love NLP").
4. Removing Special Characters: Remove special characters such as @, #, $, %, &, *, etc.
5. Reducing Duplicate Letters: Normalize repeated letters (e.g., "soooo" to "soo", "loooong" to "long").
6. Fixing Punctuation: Correct the excessive use of punctuation marks and normalize them.
7. Removing URL Artifacts: Clean up any remaining artifacts from URLs (e.g., "www.example.com////").

Submission: Submit the cleaned paragraph along with the Python code showcasing the implementation of each step. You can use popular NLP libraries like nltk, re, spaCy package.

Data: Uncleaned Paragraphs

1. OMG!! I can't believe I found this aWesoMe article about AI & machine leanring!! It was soooo gooood lol. I <3 nlp butttt i hate spelng errors in textttt. This is gonna be a looong paragraph with looooots of spacesss and weirddddd symbols @#$%. The website's link is www.example.com//// Check it out ASAP!!! #excited
2. Oh my gosh!!! Like, I can't even believe what I just stumbled upon on the world wide web. This article, "The Marvels of Artificial General Intelligence & the Future of Humanity," totally blew my mindddd!! It was, like, sooo mind-blowingly awesome, lolz. I mean, I <3 NLP butttt those annoying typos in texts drive me nuts. Brace yourselves, this is gonna be one seriously long paragraph with tons and tons of extraterrestrial spaces and some seriously weird symbols like @#$%. And guess what? The link to the website is www.incrediblenews.com//// So, um, you better check it out like ASAP!!! #excitedmuch
3. yaaayyyy!!! just watched thissss AMAZZZING vid on AI 🤖 & how it's changin' evrythinggg!! 😱 😱 it waz soooo dopeee, i can't even... lolzzz! #blessed #mindblownn... btw, the link is http://coolai.stuff.com//// totally worthhh itttttt @@@!!! i <3 tech so much but sometimes it's likee realllyyy confusinggg!!!
4. nooo waayyyyy!!!! 😳 😳 this is, like, the 1000000th time i've seen ppl mess up spellings in their blogsssss 😫 😫 i mean c'mon, u're writing abt deep learning, not sum joke topic 🙄

anywayzzzz... the article was kinda cooool butttt had lotsa weird $$$%%%@@ symbols all over. you can read it @ www.badblogger.ai//// and let me knw ur thots ASAP!!!

5. heyyyyy, did u read that crazyyyyyyy article on quantum stuff & ai?? like for realll... it wuz crazzzzy detailed but sooooo complicateddd!!! i cud barely undrstnd half of it 🤯 🤯. the diagrams were 🔥 🔥 tho!!! i found it here – www.quantmgeekzz.net//// … plz plz plz read n tell me what u thinkk 😩 😩 #toomuchinfo #brainhurts @@@!!!

Perform following experiments on above uncleaned paragraphs.

1. Load and Tokenize Text
Task: Load text from the Brown Corpus (select any category), convert it into a string, and tokenize it into individual words using NLTK.

2. Visualize Raw Word Frequencies
Task: From the tokenized words, calculate the frequency distribution and visualize it using a bar chart and a word cloud (top 20 words).

3. Apply Stemming (PorterStemmer)
Task: Apply NLTK's PorterStemmer to the tokenized words, then list each stemmed word along with its frequency.

4. Apply Lemmatization (WordNetLemmatizer)
Task: Apply NLTK's WordNetLemmatizer to the tokenized words, then list each lemmatized word along with its frequency.

5. Compare Stemming vs Lemmatization
Task: Create a side-by-side table or bar chart showing how stemming and lemmatization have reduced words differently (compare counts for at least 15 common words).

Data Cleaning — Expected Output

Paragraph 1

Cleaned text:

omg i cannot believe i found this awesome article about ai and machine learning it was soo good lol i love nlp but i hate spelling errors in text this is gonna be a long paragraph with lots of spaces and weird symbols the websites link is check it out asap excited

Paragraph 2

oh my gosh like i cannot even believe what i just stumbled upon on the world wide web this article the marvels of artificial general intelligence and the future of humanity totally blew my mind it was like soo mind blowingly awesome lol i mean i love nlp but those annoying typos in texts drive me nuts brace yourselves this is gonna be one seriously long paragraph with tons and tons of extraterrestrial spaces and some seriously weird symbols like and guess what the link to the website is so um you better check it out like asap excitedmuch

Paragraph 3

yay just watched this amazing vid on ai and how it is changing everything it was soo dope lol btw the link is totally worth it i love tech so much but sometimes it is like really confusing

Paragraph 4

no way this is like the 1000000th time i have seen people mess up spellings in their blogs i mean come on you are writing about deep learning not some joke topic anyway the article was kinda cool but had lots of weird symbols all over you can read it at and let me know your thoughts asap

Paragraph 5

hey did you read that crazy article on quantum stuff and ai like for real it was crazy detailed but soo complicated i could barely understand half of it the diagrams were fire though i found it here please please please read and tell me what you think toomuchinfo brainhurts

Marking: Marking is based on both performance during the lab hours as well as complete submission in LMS.